

End-to-End NLP Pipeline for Drug Review Analysis

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This project uses the DrugLib patient review dataset to build an end-to-end NLP workflow that handles modelling, interpretation, and retrieval in the context of healthcare drug reviews. Patient reviews contain useful information about treatment benefits, side effects, satisfaction, and condition specific concerns, but that information must be cleaned, structured, evaluated, and made searchable before it can support analysis. The project therefore connects three parts:

1. preprocessing and exploratory analysis
2. sentiment and text classification
3. BM25-based information retrieval

Dataset and preprocessing

DrugLib contains 4,143 records after combining 3,107 training reviews and 1,036 test reviews. Each review is built from three fields: `benefitsReview`, `sideEffectsReview`, and `commentsReview`. This combination is important because one field alone does not fully describe the full patient experience. Benefits capture what improved, side-effect text captures adverse reactions, and comments often explain dosage, timing, or overall satisfaction. The dataset also includes 541 unique drugs and 1,807 listed conditions, with common conditions such as depression, acne, anxiety, insomnia, and birth control.

The preprocessing notebook documents the schema, checks missing values, confirms there are no full-row duplicates, and creates model-ready text columns. The missing values are limited compared with the full dataset: 23 benefits reviews, 98 side-effect reviews, 13 comment reviews, and one missing condition. Instead of dropping large portions of data, the text fields are safely combined with missing strings handled as blanks. Ratings are converted into sentiment labels: ratings 1 to 4 are negative, 5 to 6 are neutral, and 7 to 10 are positive. This produced 2,800 positive reviews, 902 negative reviews, and 441 neutral reviews, which shows a clear class imbalance.

Text cleaning lowercases the reviews, removes formatting noise, normalizes spacing, tokenizes the text, removes standard stopwords while keeping useful negation terms, and lemmatizes tokens. The final processed corpus contains 255,156 processed tokens and a vocabulary of 13,768 terms. The average raw review length is about 119 words, while the average processed length is about 62 words. This means the reviews are detailed enough for modelling, but they also contain mixed language where benefits and side effects can appear in the same review.

Key dataset summary

Item	Result
Total records	4,143
Train / test split	3,107 train; 1,036 test
Unique drugs / conditions	541 drugs; 1,807 conditions
Sentiment split	67.6% positive; 21.8% negative; 10.6% neutral
Average review length	119 raw words; 62 processed words

Sentiment analysis

The sentiment track compares a simple lexicon method with supervised TF-IDF models. VADER was useful as a baseline because it is fast and explainable, but it was not strong enough for medical review language. It achieved 45.46% accuracy, 32.26% Macro-F1, and 45.26% Weighted-F1. The main issue is that medical reviews often contain negative health terms such as pain, nausea, anxiety, depression, or side effects even when the patient gives a positive rating because the medicine helped overall.

The first supervised model, TF-IDF with Multinomial Naive Bayes, reached 64.67% accuracy but only 26.18% Macro-F1. This result shows why accuracy alone is not reliable in an imbalanced dataset. The model leaned heavily toward the positive class and failed to meaningfully recover negative and neutral classes. The best sentiment result came from TF-IDF with Balanced Logistic Regression, which reached 74.61% accuracy, 56.41% Macro-F1, and 72.24% Weighted-F1. Its per-class F1 scores were 0.69 for negative, 0.16 for neutral, and 0.84 for positive. The neutral class remained the weakest because neutral reviews often describe both improvement and side effects.

Text classification track

The modelling track extends the sentiment work into text classification. The baseline uses term features with TF-IDF because sentiment in drug reviews is usually carried by phrases such as “worked well,” “side effects,” “waste of money,” or “helped me sleep.” A Multinomial Naive Bayes classifier with uniform class priors produced 74.8% accuracy and 67.2% Macro-F1 on a stratified split, with stronger performance on positive reviews than negative reviews. The advanced model fine-tunes a DistilBERT transformer sequence classifier with class weighting to address imbalance. Across five epochs, the transformer improved from 73.8% accuracy and 71.9% Macro-F1 in epoch 1 to about 78.3% accuracy and 75.7% Macro-F1 by epoch 5. This suggests that contextual language modelling captures review meaning better than sparse features, although it costs more training time and compute.

BM25 retrieval and search interface

The retrieval component turns the cleaned review collection into a searchable system. A custom BM25Index class tokenizes reviews, removes punctuation, lemmatizes terms, removes stopwords, computes document lengths, builds an inverted index, and scores multi-term queries using $k_1=1.2$ and $b=0.75$ by default. DrugLib-specific loading preserves metadata such as drug name, condition, rating, and effectiveness so search results are not just document IDs.

The project also includes a command-line search script. It loads the processed DrugLib CSV, builds the BM25 index, and lets a user search patient-style queries such as “anxiety insomnia sleep problems,” “nausea vomiting stomach pain,” and “birth control weight gain.” The script returns ranked reviews with drug name, condition, rating, effectiveness, score, and a shortened text snippet. For evaluation, the team created 29 condition-based queries and relevance judgments. The reported retrieval results are strong for a simple lexical system: Precision@10 = 0.679, Recall@10 = 0.138, MAP@10 = 0.561, MRR@10 = 0.869, and nDCG@10 = 0.698. Strong MRR shows that a relevant review is usually found near the top, while lower recall is expected because many relevant reviews can exist for broad conditions.

Overall findings and limitations

Overall, the project meets the main goal of connecting real-world drug review text to preprocessing, classification, model comparison, and retrieval. The strongest modelling result is the DistilBERT transformer classifier, while Balanced Logistic Regression is the best lightweight option because it gives a clear improvement over VADER and Naive Bayes without requiring heavy computation. The BM25 system is also useful because it supports direct search over patient experiences and provides explainable keyword-based results.

The main limitation is that the sentiment labels are rating-derived rather than manually annotated. A high rating can still include serious side effects, and a low rating can still mention partial benefits. Neutral reviews are especially difficult because they often contain mixed experiences. BM25 also depends on word overlap, so it can miss relevant reviews that use different wording for the same symptom or condition. Future work could add stronger medical synonym handling, a small manually reviewed sentiment sample, and a retrieval re-ranking stage so the search system can handle term mismatch more effectively.